ROOMROVER - SMART IOT BASED NAVIGATION ASSISTANCE

Group no. 15

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Introduction

RoomRover is a smart IoT-powered autonomous robot that navigates indoor spaces, detects hazards, and responds to voice queries using a local AI assistant all without relying on expensive hardware or cloud-dependent processing.

Problem Statement

Existing indoor systems lack a unified solution that combines physical mobility, hazard detection, and intelligent voice interaction in a cost-effective way.

Objective

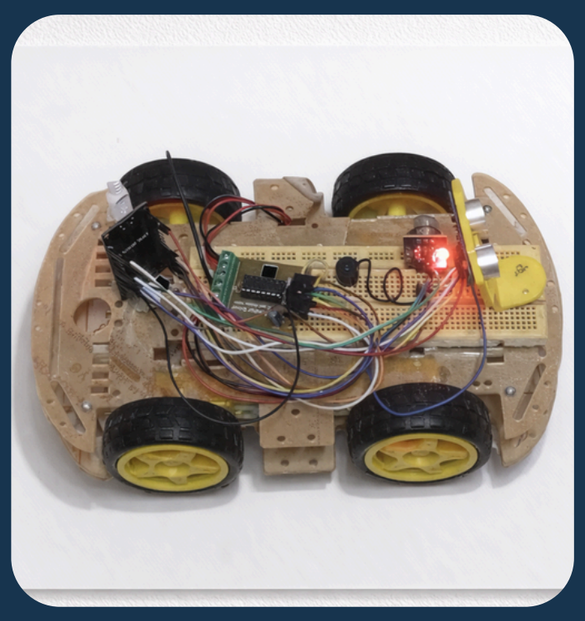
- Build an indoor navigation robot with obstacle and gas hazard detection.
- Integrate an offline AI voice assistant.
- Enable real-time remote monitoring via cloud telemetry.

Tools and Technology

Tool	Purpose
ESP32	Microcontroller
HC-SR04	Obstacle detection
MQ2	Gas/smoke sensing
L293D	Motor control
Blynk	Cloud monitoring
FAISS + GPT-2	AI voice assistant

Conclusion

RoomRover successfully combines autonomous navigation, hazard detection, and AI voice interaction into a single low-cost IoT framework, proving viable for smart indoor environments.



Future Scope

Integrate computer vision for object and person detection.

Implement SLAM-based mapping for structured indoor navigation.

Proposed Solution

RoomRover integrates an ESP32-based mobile robot with ultrasonic and MQ2 sensors for autonomous navigation and hazard detection, paired with a local RAG + Whisper voice assistant for intelligent interaction all connected via Blynk IoT cloud for real-time monitoring and alerts.

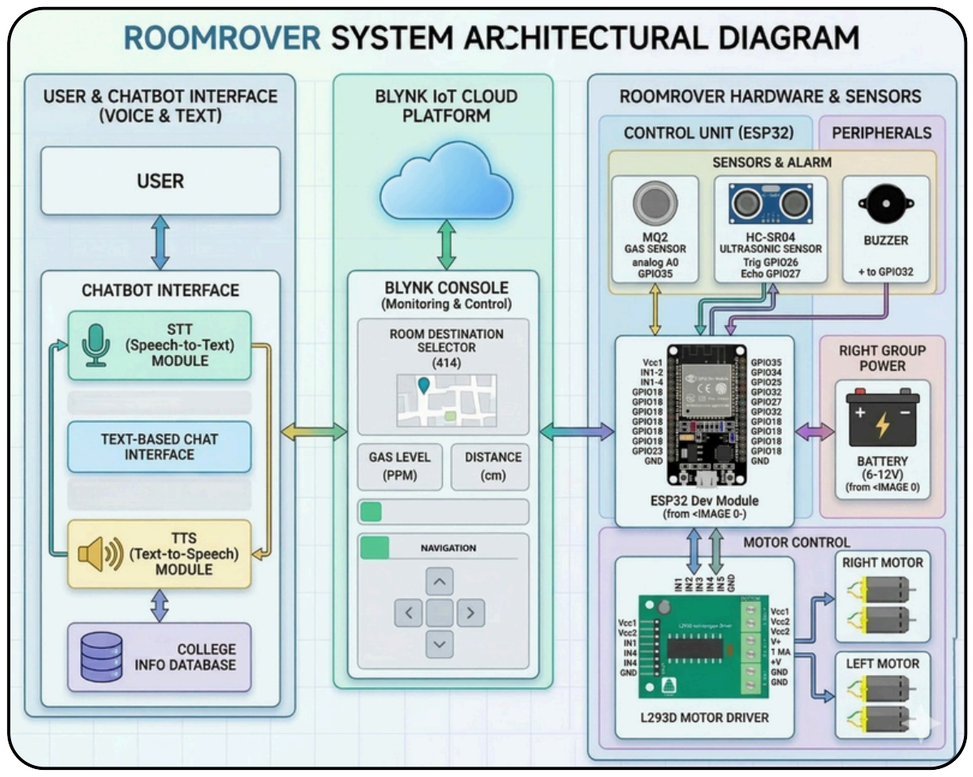


Figure 1. This figure represents the Architecture Diagram of RoomRover

Results

- Live smoke & distance data displayed on Blynk dashboard with real-time alert status.
- 100% accurate smoke alerts and reliable obstacle-based auto-halt.
- RAG chatbot delivered accurate, context-aware navigation responses.

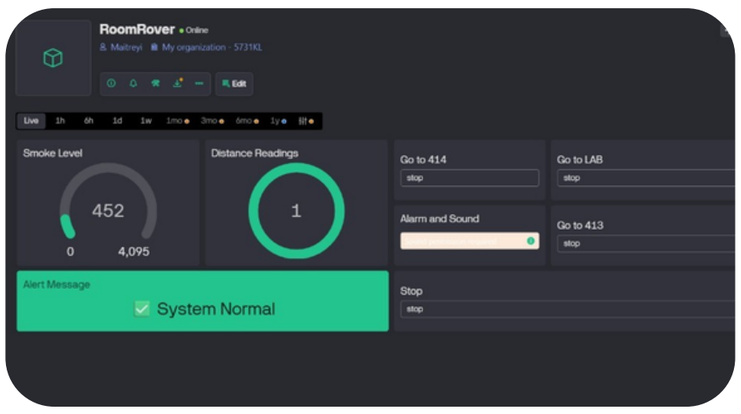


Figure 2: Blynk IoT Dashboard - live sensor readings and alert status.

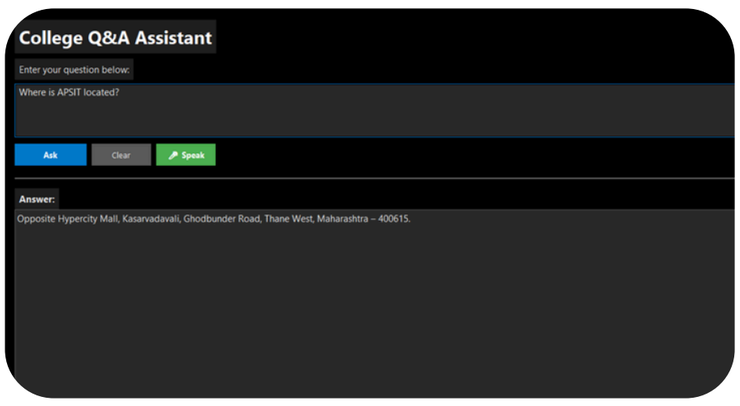


Figure 3: RoomRover Chatbot Interface - RAG-powered voice interaction.